



United International University

Department of Computer Science and Engineering

CSE 2218: Data Structures and Algorithms II Laboratory

Trimester: Fall 2023

Final Examination, Total Marks: 30, Total Time: 1 Hour 15 Minutes

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.



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Problem 1: All the roads in Krishnanagar were damaged due to a natural disaster. Raja Krishnachandra Roy, the King of Krishnanagar, is eager to reconstruct the roads. However, there isn't enough funding available to reconstruct all the roads. That's why the King wishes to rebuild the roads in a manner that connects all the sub-cities of Krishnanagar. The King is anxious about whether he can accomplish this while adhering to the specified constraints and conditions. He has asked Gopal Bhar to determine whether it is feasible.

Gopal, though not proficient in complex calculations, is determined to assist the King at any cost, as he hopes to receive Gold Coins as a reward. Since you are a renowned AlgoExpert in Krishnanagar, Gopal has sought your assistance. Help Gopal Bhar in this matter, and he pledges to share his reward with you.

You are provided with the number of sub-cities (V) and the number of roads (E) that were previously in Krishnanagar. Additionally, you are given the description of each road in the following format: (sub-city X , sub-city Y , w). Each road connects sub-cities, and the cost (w) to reconstruct each road is unique.

Furthermore, you are given the current amount of money in the fund (F) of Krishnanagar. You need to determine whether it is possible to reconstruct the roads in a way that connects all the sub-cities of Krishnanagar while staying within the available funding. If possible, print "YES"; otherwise, print "NO."

Input	Output
V E sub-city X_1 sub-city Y_1 w_1 sub-city X_2 sub-city Y_2 w_2 . . sub-city X_e sub-city Y_e w_e F	
4 5	YES

0 1 200 0 2 700 0 3 50 1 2 800 1 3 100 900	
4 5 0 1 200 0 2 700 0 3 50 1 2 800 1 3 100 500	NO



Problem 2: The King of Krishnanagar, Raja Krishnachandra Roy, has full confidence in Gopal Bhar's ability to provide an accurate decision. He plans to reward Gopal with an amount denoted as M . However, the King has set a limit and does not wish to use more than D Gold Coins, as his supply of gold coins is quite limited.

The King, mindful of the value of his time, has delegated this calculation task to you, a renowned AlgoExpert in Krishnanagar. Complying with his order, you must perform this task.

You are provided with the reward amount (M) that the King intends to give Gopal. Additionally, you are given the number of gold coins (N) that the King possesses, along with the value of each coin. Furthermore, the maximum number of Gold Coins (D) that the King is willing to use to reward Gopal is specified.

Your objective is to determine whether it is possible to give the reward while adhering to the established conditions. If it is possible, then print "YES", otherwise print "NO".

Input	Output
M N $V_1 V_2 V_3 \dots V_n$ D	
100 5 25 25 25 25 50 2	NO
100 5 25 25 25 25 50 4	YES



Don't dare to copy codes from others or talk with others. The angry Commander of Krishnanagar is looking at you.